ML-KEM.KeyGen_internal(d, z)

Uses randomness to generate an encapsulation key and a corresponding decapsulation key.

Input: randomness $d \in \mathbb{B}^{32}$.
Input: randomness $z \in \mathbb{B}^{32}$.
Output: encapsulation key $\mathsf{ek} \in \mathbb{B}^{384k+32}$.
Output: decapsulation key $\mathsf{dk} \in \mathbb{B}^{768k+96}$.

```
(ek_PKE, dk_PKE) ← K-PKE.KeyGen(d)          ▷ run key generation for K-PKE
ek ← ek_PKE                                  ▷ KEM encaps key is just the PKE encryption key
dk ← (dk_PKE ‖ ek ‖ H(ek) ‖ z)               ▷ KEM decaps key includes PKE decryption key
return (ek, dk)
```